PROJECT ACTIVITY

# Revisit Prototype 0

Use data, sources and calculator checks to prepare Prototype 1.0

1

### SPOT A WARNING SIGN

A plausible result may still contain a calculation or category error.

2

### MAP DATA AND GAPS

Separate the user's data decision from a possible calculator revision.

3

### CHECK SOURCE AND LOGIC

Trace one emission factor and run two checks through the interface.

4

### RECORD NEXT STEPS

Transfer today's notes into the Milestone 2a templates.

***Open Prototype 0 and the Shared Student Profile before the activity begins.***

PROJECT ACTIVITY

# 1. A plausible result can still be wrong

A quick check before reviewing the data

## SCENARIO

All editable activity inputs are set to zero or left blank except dorm electricity.

## DISCUSS

**What could be wrong?**

**What could you check without reading the code?**

PROJECT ACTIVITY

# 2. Data input and calculator

Decide what the user must do and what the calculator must do

### Data Input

What information does the Shared Student Profile provide?
What value can the user enter or estimate?
What assumption must be recorded?

### Calculator Design

Can Prototype 0 accept the available unit and time period?
Does it support missing information?
Does the result respond correctly to the input?

### GUIDED EXAMPLE

**A four-person dormitory used 120 kWh during the most recent 30 days.**

What should Li Ming enter in your calculator? What assumption must be recorded?
Does P0 need a revision?

The data is provided. But what if data is not provided?

PROJECT ACTIVITY

# 3. An example of data gap: hot water

Inspect how Prototype 0 supports information that the user may not know

**SHARED STUDENT PROFILE**

Li Ming takes six showers per week, about eight minutes each. The hot-water energy source and energy consumption are not provided.

1. **Find the hot-water input**
   If P0 does not include it, record that decision.
2. **Record what the calculator asks for**
   Include the unit and time period.
3. **Compare with the Shared Student Profile**
   Identify what is available and what remains unknown.
4. **Propose user support and a calculator decision**
   State one specific revision, or explain why no revision is needed.

| P0 | time period | information | Proposed treatment of the data gap | revision needed |
|---|---|---|---|---|
| **Local travel**<br>**Bus and metro distance** | km/week | PROVIDED<br>Metro: 2 one-way trips/week; 15 km/trip. No bus travel is reported. | Use 2 x 15 = 30 km/week. Walking/cycling is not added to this emitting mode. | No required change. Optional: accept trips/week and km/trip, then calculate distance. |
| **Local travel**<br>**Taxi or ride-hailing distance** | km/week | PROVIDED, DIFFERENT TIME UNIT<br>2 one-way trips/month; 10 km/trip; usually alone. | Convert: 2 x 10 x 12 / 52 = 4.6 km/week. State the annualisation rule. | Accept trips/month plus km/trip, or show a transparent converter. Do not silently retain the 5 km/week default. |
| **Long-distance travel**<br>**Intercity rail distance** | km/year | PARTIAL FOR ANNUAL ENTRY<br>2 round trips/teaching semester; 220 km each way. Semesters/year is not stated in the profile. | If the group assumes 2 teaching semesters/year: 2 x 2 x 2 x 220 = 1,760 km/year. Record the assumption. | Accept round trips/semester and one-way distance, or show the conversion and editable semesters/year. |
| **… …** | | | | |
| **Waste**<br>**Mixed waste** | kg/week on campus | PARTIAL<br>About 3 small bags/week. Mass and treatment route are unknown. | Use bags/week with a documented editable kg/bag proxy. If no defensible estimate is possible, allow N/A/skip.. | Accept bag count plus kg/bag and, if available, treatment route. Do not retain 2 kg/week as a fact. |
| **Food**<br>**Meals containing meat** | meals/week | PROVIDED FOR MEAT COUNT<br>3 beef/lamb + 8 other meat = 11. The other 10 meals include 5 fish/egg/dairy/soy and 5 mainly plant based. | Enter 11 for the present input, but flag that P0 treats all remaining meals as plant based. | Revise food inputs to represent the profile categories, or clearly explain the simplified grouping. |

PROJECT ACTIVITY

# 4. Emission factor source check

Follow one factor from Prototype 0 to an identifiable original source

**1**

**FACTOR IN P0**

Record the value and unit.

**2**

**SOURCE STATED**

Does P0 identify a source?

**3**

**ORIGINAL SOURCE**

Can you locate the source, e.g., open the document or webpage?

**4**

**VALUE AND UNIT**

Do they match the source?

**5**

**DECISION**

Keep, replace, or cannot verify yet.

***AI may suggest where to look. The original source provides the evidence.***

After class: document three factors from at least two categories.

| Activity and category | Factor currently used | Source check | Fit with this calculation | Decision |
|---|---|---|---|---|
| **EF 1**<br>**Dorm electricity**<br>**Category: Dorm energy** | 0.60 kg CO2e/kWh | Source shown in P0: NO<br>P0 labels the value illustrative and uncited. Exact original source: NOT LOCATABLE.<br>Official replacement source located: 上海市生态环境局_上海市统计局关于发布2020-2024年度上海市公共电网电力、热力二氧化碳排放因子的通知_其他文件_上海市生态环境局 | Unit: compatible with kWh input.<br>Geography/year: Shanghai China, 2024. | REPLACE<br>Use 0.4701 kg CO2e/kWh for this worked example. Record source, year, geography boundary; keep the value editable. |
| | | | | |
| | | | | |

PROJECT ACTIVITY

# 5. Calculator logic checks

Use the interface to identify calculation or category errors

CHECK A

## Single activity

1. Set every editable activity input to zero or blank, as appropriate.

2. Enter a value for dorm electricity only.

3. Check which categories report emissions.

CHECK B

## Change one input

1. Keep every other input unchanged.

2. Increase or decrease the electricity value.

3. Check what changes and what remains unchanged.

**Record: expected result, observed result, and revision needed**

PROJECT ACTIVITY

AFTER CLASS

## 6. Continue the review after class

Complete the review before asking AI to develop Prototype 1.0

### MILESTONE 2a

- Map every activity input
- Plan support for partial or missing data
- Review three factors from at least two categories
- Complete the calculator logic checks
- Prepare the Revision Brief and AI-Aided Log

**Submit two completed PDF templates**

### MILESTONE 2b

**Upload the complete Prototype 1.0 HTML file.**

Prototype 1.0 is submitted here, not under Milestone 2a.

### MILESTONE 2c

**Submit an individual reflection of 100 to 180 words.**

Optional: use an AI coding tool such as Trae. Tool choice does not affect assessment.